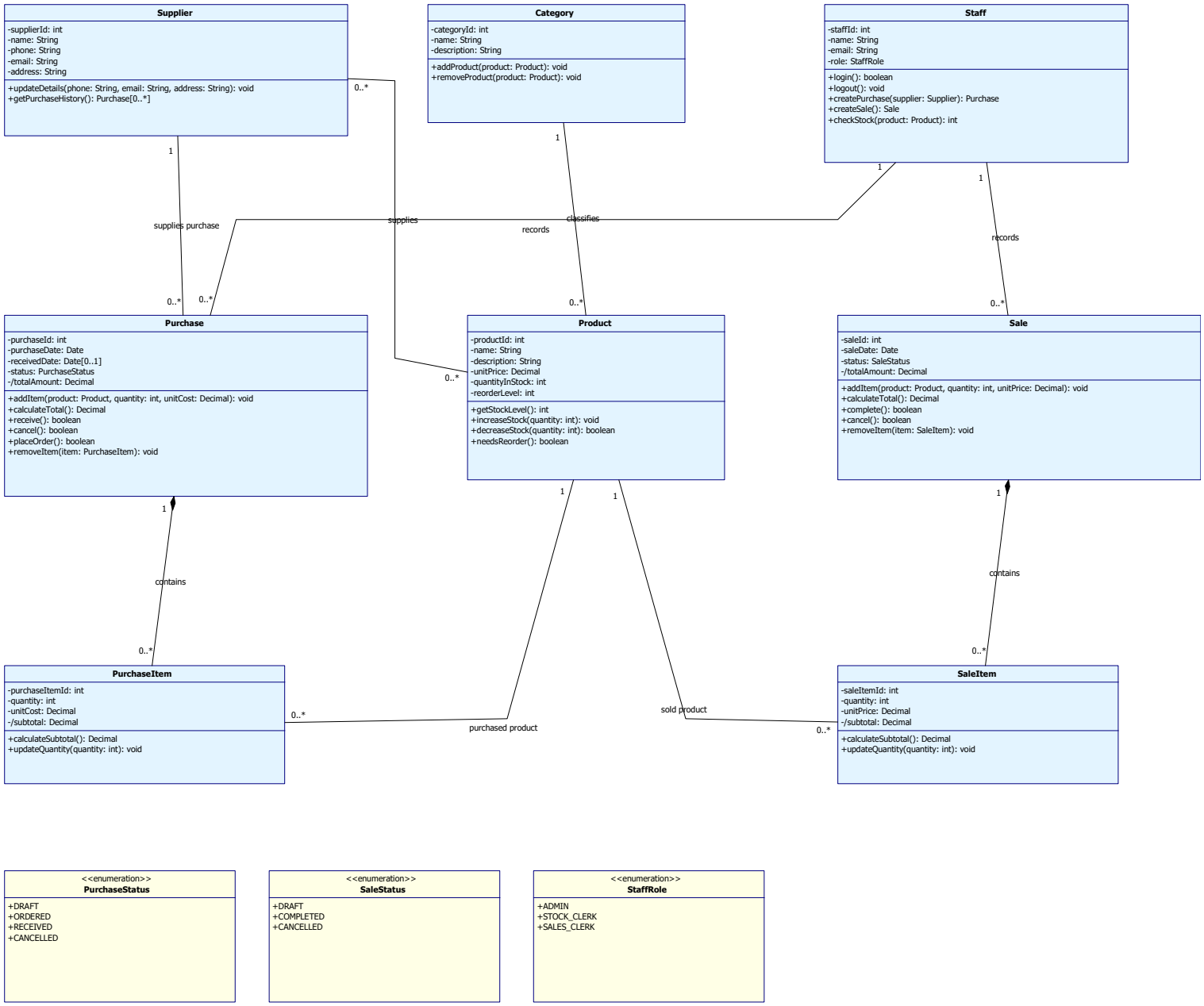

Supplier
-supplierId: int
-name: String
-phone: String
-email: String
-address: String
+updateDetails(phone: String, email: String, address: String): void
+getPurchaseHistory(): Purchase[0..*]
Category
-categoryId: int
-name: String
-description: String
+addProduct(product: Product): void
+removeProduct(product: Product): void
Staff
-staffId: int
-name: String
-email: String
-role: StaffRole
+login(): boolean
+logout(): void
+createPurchase(supplier: Supplier): Purchase
+createSale(): Sale
+checkStock(product: Product): int
Purchase
-purchaseId: int
-purchaseDate: Date
-receivedDate: Date[0..1]
-status: PurchaseStatus
-/totalAmount: Decimal
+addItem(product: Product, quantity: int, unitCost: Decimal): void
+calculateTotal(): Decimal
+receive(): boolean
+cancel(): boolean
+placeOrder(): boolean
+removeItem(item: PurchaseItem): void
Product
-productId: int
-name: String
-description: String
-unitPrice: Decimal
-quantityInStock: int
-reorderLevel: int
+getStockLevel(): int
+increaseStock(quantity: int): void
+decreaseStock(quantity: int): boolean
+needsReorder(): boolean
Sale
-saleId: int
-saleDate: Date
-status: SaleStatus
-/totalAmount: Decimal
+addItem(product: Product, quantity: int, unitPrice: Decimal): void
+calculateTotal(): Decimal
+complete(): boolean
+cancel(): boolean
+removeItem(item: SaleItem): void
PurchaseItem
-purchaseItemId: int
-quantity: int
-unitCost: Decimal
-/subtotal: Decimal
+calculateSubtotal(): Decimal
+updateQuantity(quantity: int): void
SaleItem
-saleItemId: int
-quantity: int
-unitPrice: Decimal
-/subtotal: Decimal
+calculateSubtotal(): Decimal
+updateQuantity(quantity: int): void
supplies purchase
supplies
records
classifies
records
contains
contains
purchased product
sold product
1
0..*
<<enumeration>>
PurchaseStatus
+DRAFT
+ORDERED
+RECEIVED
+CANCELLED
<<enumeration>>
SaleStatus
+DRAFT
+COMPLETED
+CANCELLED
<<enumeration>>
StaffRole
+ADMIN
+STOCK_CLERK
+SALES_CLERK